

NEET BIOLOGY

Practice Question Paper | Botany & Zoology

Questions: 50

Max Marks: 200

Duration: 60 min

Marking: +4 / -1

General Instructions

- Each correct answer carries **+4 marks**. Each wrong answer carries **-1 mark**. Unattempted questions carry 0 marks.
- Only one option is correct for each question.
- Section A - Botany:** Questions 1-25 | **Section B - Zoology:** Questions 26-50
- Answers with explanations are provided at the end of this paper.

Section A - Botany (Q. 1-25)

Q.1 Which of the following is the correct sequence of events during meiosis I? Cell Division

- | | |
|--|--|
| A Leptotene - Zygotene - Diplotene - Pachytene - Diakinesis | B Leptotene - Zygotene - Pachytene - Diplotene - Diakinesis |
| C Zygotene - Leptotene - Pachytene - Diplotene - Diakinesis | D Leptotene - Pachytene - Zygotene - Diplotene - Diakinesis |

Q.2 In which of the following plants is the stem modified into a flattened, leaf-like structure called phylloclad? Morphology

- | | |
|------------------|--------------------|
| A Opuntia | B Asparagus |
| C Ruscus | D Euphorbia |

Q.3 The process by which glucose is oxidised to pyruvate in the cytoplasm is called: Respiration

- | | |
|------------------------------------|-------------------------|
| A Glycolysis | B Krebs cycle |
| C Oxidative phosphorylation | D Beta-oxidation |

Q.4 Which organelle is called the 'powerhouse of the cell' and has its own circular DNA? Cell Biology

- | | |
|-------------------|-----------------------|
| A Nucleus | B Chloroplast |
| C Ribosome | D Mitochondria |

Q.5 Cycas belongs to which division of the plant kingdom? Plant Kingdom

- | | |
|-----------------------|-----------------------|
| A Bryophyta | B Gymnospermae |
| C Angiospermae | D Pteridophyta |

Q.6	Which pigment absorbs light most efficiently at wavelengths of 430 nm and 680 nm during photosynthesis?	<i>Photosynthesis</i>
A	Chlorophyll b	B Xanthophyll
C	Chlorophyll a	D Carotene
Q.7	The 'fluid mosaic model' of cell membrane was proposed by:	<i>Cell Biology</i>
A	Singer and Nicolson	B Watson and Crick
C	Schleiden and Schwann	D Davson and Danielli
Q.8	Which of the following is a characteristic of C4 plants?	<i>Photosynthesis</i>
A	First stable product is 3-PGA	B High CO ₂ compensation point
C	Absence of bundle sheath cells	D Kranz anatomy is present
Q.9	In angiosperms, double fertilisation involves:	<i>Reproduction</i>
A	One sperm fuses with egg, another with antipodals	B One sperm fuses with egg, another with polar nuclei
C	Both sperms fuse with polar nuclei	D One sperm fuses with synergid, another with egg
Q.10	Which enzyme is responsible for the formation of phosphodiester bonds during DNA replication?	<i>Molecular Biology</i>
A	DNA helicase	B DNA primase
C	DNA polymerase III	D Topoisomerase
Q.11	Vernalisation is the requirement of _____ for flowering in plants.	<i>Plant Physiology</i>
A	Low temperature treatment	B High temperature treatment
C	Short photoperiod	D Long photoperiod
Q.12	Which of the following is NOT a function of auxin (IAA)?	<i>Plant Hormones</i>
A	Apical dominance	B Cell elongation
C	Root initiation	D Stomatal closure
Q.13	The number of ATP molecules produced by complete oxidation of one glucose molecule through aerobic respiration is:	<i>Respiration</i>
A	36 ATP	B 38 ATP
C	30 ATP	D 32 ATP

Q.14	Casparian strips are found in:	<i>Plant Anatomy</i>
A	Epidermis	B Cortex
C	Endodermis	D Pericycle

Q.15	The law of independent assortment does NOT hold when the two genes are:	<i>Genetics</i>
A	Linked on the same chromosome	B Located on different chromosomes
C	Both dominant	D Showing codominance

Q.16	Which of the following is a free-living, nitrogen-fixing bacterium found in soil?	<i>Ecology</i>
A	Rhizobium	B Nitrosomonas
C	Pseudomonas	D Azotobacter

Q.17	The primary acceptor of CO ₂ in C ₃ plants is:	<i>Photosynthesis</i>
A	OAA (Oxaloacetic acid)	B RuBP (Ribulose biphosphate)
C	PEP (Phosphoenolpyruvate)	D Pyruvate

Q.18	Which statement about the cohesion-tension theory of ascent of sap is CORRECT?	<i>Plant Physiology</i>
A	It involves active transport through phloem	B Root pressure alone drives water to the top
C	Transpiration pull creates tension in xylem column	D It was proposed by Munch

Q.19	Which of the following is an example of a plant with epipetalous stamens?	<i>Morphology</i>
A	Brinjal (Solanum melongena)	B China rose (Hibiscus rosa-sinensis)
C	Pea (Pisum sativum)	D Lotus (Nelumbo nucifera)

Q.20	Geitonogamy is:	<i>Reproduction</i>
A	Transfer of pollen from one plant to another plant	B Transfer of pollen within the same flower
C	Pollination by insects	D Transfer of pollen between different flowers of the same plant

Q.21	The technique of PCR (Polymerase Chain Reaction) was developed by:	<i>Biotechnology</i>
A	Stanley Cohen	B Kary Mullis
C	Herbert Boyer	D Frederick Sanger

Q.22	Which hormone promotes dormancy in seeds and causes wilting of leaves?	<i>Plant Hormones</i>
A	Gibberellin	B Cytokinin
C	Abscisic Acid (ABA)	D Ethylene

Q.23	In a typical dicot root, the vascular bundles are:	<i>Plant Anatomy</i>
A	Radial and exarch xylem	B Collateral and endarch xylem
C	Bicollateral and mesarch xylem	D Conjoint and endarch xylem

Q.24	Hardy-Weinberg principle states that allele frequency in a population remains constant if:	<i>Evolution</i>
A	Natural selection is acting	B Genetic drift is occurring
C	Migration is happening	D Random mating, no selection, no mutation, no migration

Q.25	Which is the CORRECT ploidy level of endosperm in a typical angiosperm?	<i>Reproduction</i>
A	Diploid (2n)	B Triploid (3n)
C	Haploid (n)	D Tetraploid (4n)

Section B - Zoology (Q. 26-50)

Q.26	Which of the following is an acoelomate organism?	<i>Animal Kingdom</i>
A	Earthworm	B Roundworm
C	Planaria	D Starfish

Q.27	The sliding filament theory of muscle contraction involves movement of:	<i>Locomotion</i>
A	Actin filaments over myosin	B Myosin filaments over actin
C	Z-line towards M-line	D H-zone expanding during contraction

Q.28	The pacemaker of the human heart is:	<i>Circulation</i>
A	AV node	B Bundle of His
C	Purkinje fibres	D SA node

Q.29	Haemoglobin is a:	<i>Biomolecules</i>
A	Monomeric protein with one haem group	B Tetrameric protein with four haem groups
C	Dimeric protein with two haem groups	D Trimeric protein with three haem groups

Q.30	In the human kidney, ultrafiltration takes place in the:	<i>Excretion</i>
A	Loop of Henle	B Proximal convoluted tubule
C	Malpighian corpuscle (Bowman's capsule)	D Collecting duct

Q.31	Which hormone is secreted by beta-cells of Islets of Langerhans?	<i>Chemical Coordination</i>
A	Insulin	B Glucagon
C	Somatostatin	D Gastrin

Q.32	The resting membrane potential of a neuron is approximately:	<i>Neural Control</i>
A	+70 mV	B -40 mV
C	+40 mV	D -70 mV

Q.33	Which blood group is called 'Universal Donor' and 'Universal Recipient' respectively?	<i>Body Fluids</i>
A	AB and O	B O and AB
C	A and B	D B and A

Q.34	Spermatogenesis occurs in the:	<i>Reproduction</i>
A	Epididymis	B Rete testis
C	Seminiferous tubules	D Vas deferens

Q.35	Which of the following is the CORRECT match of the digestive enzyme and its substrate?	<i>Digestion</i>
A	Lipase - Fats	B Amylase - Proteins
C	Trypsin - Lipids	D Pepsin - Carbohydrates

Q.36	The functional unit of a kidney is:	<i>Excretion</i>
A	Glomerulus	B Bowman's capsule
C	Collecting duct	D Nephron

Q.37	Transcription in eukaryotes takes place in the:	<i>Molecular Biology</i>
A	Cytoplasm	B Nucleus
C	Mitochondria only	D Ribosome

Q.38	Which type of immunity involves T-lymphocytes and is cell-mediated?	<i>Human Health</i>
A	Humoral immunity	B Passive immunity
C	Cell-mediated immunity (CMI)	D Innate immunity

Q.39	During oogenesis, which stage is present at birth in the human female?	<i>Reproduction</i>
A	Primary oocyte arrested at prophase I	B Secondary oocyte at metaphase II
C	Oogonia undergoing mitosis	D Mature ovum

Q.40	Rohu (Labeo rohita) belongs to which class of vertebrates?	<i>Animal Kingdom</i>
A	Amphibia	B Cyclostomata
C	Chondrichthyes	D Osteichthyes

Q.41	Which of the following is the 'master gland' of the endocrine system?	<i>Chemical Coordination</i>
A	Thyroid	B Anterior pituitary
C	Hypothalamus	D Adrenal gland

Q.42	Which of the following sexually transmitted infections (STIs) is caused by a virus?	<i>Human Health</i>
A	Gonorrhoea	B Syphilis
C	Genital herpes	D Chlamydia

Q.43	The one gene-one enzyme hypothesis was proposed by:	<i>Genetics</i>
A	Beadle and Tatum	B Morgan and Bridges
C	Watson and Crick	D Garrod and Bateson

Q.44	Which part of the brain controls body temperature, hunger, and thirst?	<i>Neural Control</i>
A	Cerebrum	B Cerebellum
C	Medulla oblongata	D Hypothalamus

Q.45	In Down's syndrome, the karyotype is:	<i>Genetics</i>
A	45 chromosomes (monosomy 21)	B 47 chromosomes with trisomy 21
C	47 chromosomes with XXY	D 45 chromosomes with XO

Q.46	Which type of ecological pyramid is always upright and can never be inverted?	<i>Ecology</i>
A	Pyramid of biomass	B Pyramid of numbers
C	Pyramid of energy	D Both A and B

Q.47	In which phase of the cell cycle does DNA replication occur?	<i>Cell Division</i>
A	S phase	B G1 phase
C	G2 phase	D M phase

Q.48	Which of the following is a 'test cross'?	<i>Genetics</i>
A	Crossing two F1 individuals	B Crossing two homozygous dominant plants
C	Crossing F1 with homozygous dominant	D Crossing F1 with homozygous recessive

Q.49	Insulin is composed of:		<i>Chemical Coordination</i>
A	One polypeptide chain with 21 amino acids	B	Two polypeptide chains A (21 AA) and B (30 AA) linked by disulfide bonds
C	Three polypeptide chains linked by hydrogen bonds	D	A single chain of 51 amino acids with no disulfide bonds

Q.50	Which of the following biomes has the HIGHEST biodiversity?		<i>Ecology</i>
A	Temperate deciduous forest	B	Tundra
C	Tropical rainforest	D	Grassland

Answer Key & Explanations

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	B	11	A	21	B	31	A	41	B
2	C	12	D	22	C	32	D	42	C
3	A	13	B	23	A	33	B	43	A
4	D	14	C	24	D	34	C	44	D
5	B	15	A	25	B	35	A	45	B
6	C	16	D	26	C	36	D	46	C
7	A	17	B	27	A	37	B	47	A
8	D	18	C	28	D	38	C	48	D
9	B	19	A	29	B	39	A	49	B
10	C	20	D	30	C	40	D	50	C

Detailed Explanations

Section A - Botany

Q.1 (B) Leptotene - Zygotene - Pachytene - Diplotene - Diakinesis

Correct sequence of sub-stages of prophase I: Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis.

Q.2 (C) Ruscus

Ruscus has a flattened phylloclad. Opuntia has a fleshy cladode; Asparagus has needle-like cladodes.

Q.3 (A) Glycolysis

Glycolysis occurs in the cytoplasm and converts 1 glucose into 2 pyruvate molecules, yielding a net 2 ATP.

Q.4 (D) Mitochondria

Mitochondria is the powerhouse of the cell. It has circular DNA and 70S ribosomes, supporting the endosymbiotic theory.

Q.5 (B) Gymnospermae

Cycas is a gymnosperm - a primitive seed-bearing plant with naked seeds not enclosed in fruit.

Q.6 (C) Chlorophyll a

Chlorophyll a absorbs maximally at ~430 nm (blue-violet) and ~680 nm (red). It is the primary photosynthetic pigment.

Q.7 (A) Singer and Nicolson

Singer and Nicolson (1972) proposed the fluid mosaic model, explaining the dynamic nature of the plasma membrane.

Q.8 (D) Kranz anatomy is present

C4 plants show Kranz anatomy with well-developed bundle sheath cells. First stable product is OAA (4C).

Q.9 (B) One sperm fuses with egg, another with polar nuclei

In double fertilization: one sperm + egg = zygote (2n); one sperm + 2 polar nuclei = primary endosperm nucleus (3n).

Q.10 (C) DNA polymerase III

DNA polymerase III (prokaryotes) / DNA polymerase delta (eukaryotes) catalyzes phosphodiester bond formation.

Q.11 (A) Low temperature treatment

Vernalisation is the induction of flowering in plants by exposure to prolonged cold (low temperature). E.g., winter wheat.

Q.12 (D) Stomatal closure

Stomatal closure is a function of abscisic acid (ABA), not auxin. Auxin promotes cell elongation, apical dominance, and root formation.

Q.13 (B) 38 ATP

Theoretical yield is 38 ATP per glucose in prokaryotes (36-38 in eukaryotes depending on shuttle mechanism used).

Q.14 (C) Endodermis

Casparian strips are suberin thickenings in endodermal cell walls. They regulate apoplastic transport of water and minerals.

Q.15 (A) Linked on the same chromosome

Linked genes (on same chromosome) do not assort independently; they tend to be inherited together, violating Mendel's 2nd law.

Q.16 (D) Azotobacter

Azotobacter is a free-living aerobic nitrogen fixer. Rhizobium is symbiotic. Nitrosomonas oxidizes NH_3 to NO_2^- (nitrification).

Q.17 (B) RuBP (Ribulose biphosphate)

In C3 plants (Calvin cycle), CO_2 is accepted by RuBP (5C) to form 2 molecules of 3-PGA (3C). RuBisCO catalyzes this reaction.

Q.18 (C) Transpiration pull creates tension in xylem column

Dixon and Joly proposed cohesion-tension theory. Transpiration pull creates tension; cohesion holds the water column.

Q.19 (A) Brinjal (Solanum melongena)

In brinjal (Solanaceae), stamens are epipetalous (attached to petals). Hibiscus has epiphyllous, monadelphous stamens.

- Q.20 (D) Transfer of pollen between different flowers of the same plant**
Geitonogamy is pollen transfer from one flower to another on the same plant. Functionally cross-pollination but genetically self-pollination.
- Q.21 (B) Kary Mullis**
Kary Mullis developed PCR in 1983, for which he received the Nobel Prize in Chemistry in 1993.
- Q.22 (C) Abscissic Acid (ABA)**
ABA is the stress hormone. It promotes seed dormancy, causes stomatal closure (wilting), and inhibits growth.
- Q.23 (A) Radial and exarch xylem**
In dicot roots, vascular bundles are radial (xylem and phloem alternate), and xylem is exarch (protoxylem towards periphery).
- Q.24 (D) Random mating, no selection, no mutation, no migration**
H-W equilibrium requires: large population, random mating, no mutation, no natural selection, no gene flow.
- Q.25 (B) Triploid (3n)**
Endosperm is 3n (triploid), formed by fusion of one sperm (n) + two polar nuclei (n+n) during double fertilization.

Section B - Zoology

- Q.26 (C) Planaria**
Planaria (Platyhelminthes) is acoelomate (no body cavity). Roundworm is pseudocoelomate; Earthworm and Starfish are true coelomates.
- Q.27 (A) Actin filaments over myosin**
Actin (thin) filaments slide over myosin (thick) filaments. Z-lines move towards each other; H-zone and I-band shorten; A-band remains constant.
- Q.28 (D) SA node**
SA node (sinoatrial node) in the right atrium is the natural pacemaker, generating impulses at 70-75 beats/minute.
- Q.29 (B) Tetrameric protein with four haem groups**
Haemoglobin is a tetrameric protein (2alpha + 2beta subunits), each with one haem group. It can carry up to 4 O₂ molecules.
- Q.30 (C) Malpighian corpuscle (Bowman's capsule)**
Ultrafiltration occurs in the Malpighian corpuscle (glomerulus + Bowman's capsule). Blood pressure forces small molecules into Bowman's capsule.
- Q.31 (A) Insulin**
Beta-cells secrete insulin (lowers blood glucose). Alpha-cells secrete glucagon (raises blood glucose). Delta-cells secrete somatostatin.

Q.32	(D) -70 mV Resting membrane potential is about -70 mV (inside negative), maintained by Na ⁺ /K ⁺ ATPase pump.
Q.33	(B) O and AB Blood group O (Rh-) is universal donor (no antigens on RBCs). Blood group AB is universal recipient (no antibodies in plasma).
Q.34	(C) Seminiferous tubules Spermatogenesis occurs in the seminiferous tubules of the testis. Sertoli cells support developing spermatids; Leydig cells produce testosterone.
Q.35	(A) Lipase - Fats Lipase digests fats into fatty acids and glycerol. Amylase acts on starch. Trypsin and Pepsin are protease enzymes acting on proteins.
Q.36	(D) Nephron Nephron is the structural and functional unit of the kidney. Each human kidney has about 1 million nephrons.
Q.37	(B) Nucleus In eukaryotes, transcription occurs in the nucleus. The mRNA is processed (capping, tailing, splicing) before export to cytoplasm for translation.
Q.38	(C) Cell-mediated immunity (CMI) Cell-mediated immunity (CMI) involves T-lymphocytes; responsible for graft rejection and killing virus-infected cells.
Q.39	(A) Primary oocyte arrested at prophase I Female babies are born with ~2 million primary oocytes arrested at prophase I of meiosis I (diplotene stage).
Q.40	(D) Osteichthyes Labeo rohita (Rohu) is a bony fish belonging to class Osteichthyes. Sharks are Chondrichthyes. Lamprey is Cyclostomata.
Q.41	(B) Anterior pituitary Anterior pituitary is the 'master gland' - it regulates most other endocrine glands through tropic hormones (TSH, ACTH, FSH, LH, GH, Prolactin).
Q.42	(C) Genital herpes Genital herpes is caused by Herpes simplex virus (HSV). Gonorrhoea, Syphilis, and Chlamydiasis are bacterial infections.
Q.43	(A) Beadle and Tatum Beadle and Tatum (1941) proposed the one gene-one enzyme hypothesis using Neurospora crassa. They won the Nobel Prize in 1958.

Q.44 (D) Hypothalamus

Hypothalamus acts as the 'thermostat'. It regulates body temperature, hunger, thirst, and controls pituitary hormone release.

Q.45 (B) 47 chromosomes with trisomy 21

Down's syndrome is trisomy 21 ($2n+1 = 47$). XXY is Klinefelter's syndrome. XO (45) is Turner's syndrome.

Q.46 (C) Pyramid of energy

Pyramid of energy is always upright because energy is lost at each trophic level (10% law). Pyramids of numbers and biomass can be inverted.

Q.47 (A) S phase

DNA replication occurs during S (Synthesis) phase of the cell cycle, between G1 and G2. DNA content doubles from 2C to 4C.

Q.48 (D) Crossing F1 with homozygous recessive

A test cross involves crossing an individual of unknown genotype (or F1) with the homozygous recessive parent to determine genotype.

Q.49 (B) Two polypeptide chains A (21 AA) and B (30 AA) linked by disulfide bonds

Insulin consists of chain A (21 amino acids) and chain B (30 amino acids) connected by two disulfide bonds. It is derived from proinsulin.

Q.50 (C) Tropical rainforest

Tropical rainforests have the highest species richness. They cover ~7% of Earth's surface but harbour over 50% of all species.